Anonymisation of quantitative data

EUR Data Coffee Breaks

April 11 & 22, 2024

Katia Roque,
Research Data Steward, ESHPM

Erasmus University Rotterdam

EUR Data Coffee Breaks

- Monthly meetings about Research Data

  Organized by the EUR Data Stewards
- Short, focused, and informal
- 15-minute presentations
- Twice a month (online)
- No recordings, slides available on EUR Data Stewardship (Zenodo)

EUR Data Coffee Breaks

Anonymisation of quantitative data

Program for today:

Introduction: why is anonymisation needed

Key Concepts

Difference between Anonymisation and Pseudonymisation

How to anonymise quantitative data?

Anonymisation techniques

Further Resources and questions

Introduction: Why is anonymisation needed

Privacy and the need to protect the data and the confidentiality of participants:

GDPR

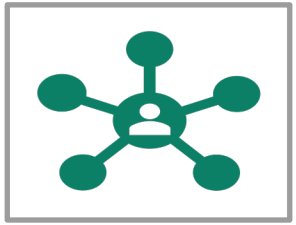
Anonymous data is no longer considered personal data

Open Science:

Making data shareable and reusable

Meeting requirements from funding bodies

Key Concepts: Direct Identifiers



Information collected by the researcher(s) that can be directly linked to a participant

Examples:

Names

Addresses

Email addresses

Telephone numbers

BSN

Detailed geographical location

IP addresses

Exact dates that can be easily linked to an individual

Key Concepts: Indirect Identifiers

- Characteristics that when cross-referenced with other sources can be traced back to an individual

These can be demographic or social information:

- Age
- Gender
- Occupational information
- Education history
- Workplace
- Income
- Social connections
- Geographic or other unique features of a community

Key Concepts: data minimisation

- Refers to the process of gathering the least amount of data essential for conducting research and achieving the intended research objectives (e.g., answering the research question).

Data Collection

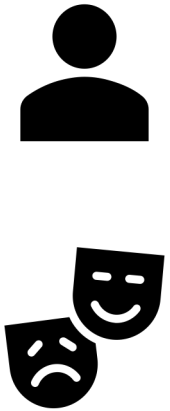
Data minimisation

Anonymisation vs Pseudonymisation

- Anonymised data:
  - Data is rendered unidentifiable even by the researcher who collected the data (consultation with a privacy officer can help you determine whether your data can be considered fully anonymous)
  - Thus, when data is fully anonymous, it is not possible to identify an individual through direct or indirect identifiers
- Complete or full anonymisation is not always possible
- Consent, anonymisation and access controls

Anonymisation vs Pseudonymisation

- Anonymized data is no longer bound by GDPR regulations
- Pseudonymous data is still personal data.



Anonymisation vs Pseudonymisation

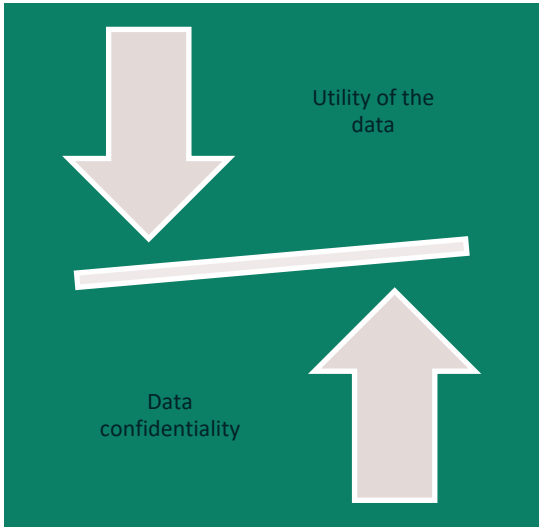
Higher loss of information

Anonymised

De-identified (including pseudonymised)

Higher utility of the data

Non-anonymized data



In general, anonymisation of quantitative data is easier than anonymisation of qualitative data.

How to anonymise quantitative data?

Practical steps:

Identify direct identifiers in the dataset.

Remove, aggregate or redact them

Recognize indirect identifiers, check for outliers or sensitive information.

Maintain consistency throughout the anonymisation process.

Once anonymisation is complete, delete the original data files.

Review background materials related to the data

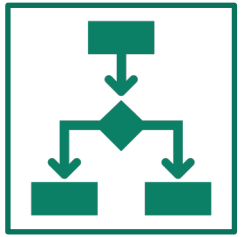
Erase or anonymize any identifiable information found in the background materials.

Anonymisation techniques

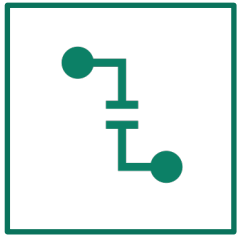
- Removing variables, values and units of observation
- Aggregating or recoding variable values
- Editing responses in open-ended variables
- K-anonymity and l-diversity
- Noise addition
- Permutation

Removing variables, values and units of observation

Removing direct or indirect identifiers to decrease the risk of identification



Open-ended variables may need removal, especially if they duplicate other data



If data from open-ended variable is needed, separate open-ended data into a different file to prevent linking.

Aggregating or recoding

- Preferable to simply removing variables
- Aggregate or reduce the precision of variables (birth date- birth year- year range)
- Recode categorical variables into fewer categories

Original Data	Aggregate data
School name: Rotterdam International School	School type: Secondary School
Age: 14	Age range: 10-18
City/Region: Rotterdam	Country: Netherlands

Editing responses in open-ended variables

⚠ Asses the risk of disclosure

Generalize meaning of text variables- replace potentially disclosive text with more general text

Special attention to unique information that could reveal identities, in these situations removing or replacing a variable might not be enough.

Example:
Occupation -> [a starting position]
Workplace -> [a construction company]
Eye surgery-> [medical procedure]

K-anonymity and l-diversity

- **K-anonymity-** mathematical way to show that a dataset has been anonymised
- involves grouping data units with similar attributes to prevent individual identification
- k- anonymisation alone may not prevent disclosure, can be accompanied by l-diversity
- L-diversity ensures variability in sensitive attributes within each group

	Race	Birth	Gender	ZIP	Problem
t1	Black	1965	m	0214*	short breath
t2	Black	1965	m	0214*	chest pain
t3	Black	1965	f	0213*	hypertension
t4	Black	1965	f	0213*	hypertension
t5	Black	1964	f	0213*	obesity
t6	Black	1964	f	0213*	chest pain
t7	White	1964	m	0213*	chest pain
t8	White	1964	m	0213*	obesity
t9	White	1964	m	0213*	short breath
t10	White	1967	m	0213*	chest pain
t11	White	1967	m	0213*	chest pain

k-Anonymity and cluster- based methods for privacy: https://elf11.github.io/2017/04/22/kanonymity.html (accessed April 7,2024)available under Creative Commons Attribution 4.0 License

Noise addition

- Introducing noise involves tweaking data attributes to decrease their accuracy,

  Noise can be introduced through various methods:

- For instance, age attribute values could be presented with an accuracy range of +/-2 years:

  Age 34 -> Age 36

- Multiplying original values by random numbers or transforming categorized values.

  Example: Transforming 10% of Leiden residents to residents of The Hague.

- Aggregation of identifiable continuous variable values into group means

  Example: Replacing exact medication costs for patients with a serious illness with group mean costs.

Permutation

- Permutation involves swapping values of variables containing indirect identifiers.
- It is like shuffling cards - it changes the order of information without changing its overall shape.
- Useful when information isn't strongly correlated, but not so great when there are strong correlations.
- Mixing permutation with other methods helps keep things safe.

Questions

Acknowledgements

- Annika Valaranta. (2021). How to guide anonymisation?. Global Virtual Conference (IASSIST 2021), Online. Zenodo. https://doi.org/10.5281/zenodo.5181069
- E. Glerean. (2024). Basics of personal data anonymisation. Zenodo. https://doi.org/10.5281/zenodo.10649060
- Finnish Social Science Data Archive (FSD) (n.d.). *Anonymisation and Personal Data* https://www.fsd.tuni.fi/en/services/data-management-guidelines/anonymisation-and-identifiers/#anonymisation-of-quantitative-data
- Joanna Mania. (2023). Anonymization of qualitative data – a practical approach. Zenodo. https://doi.org/10.5281/zenodo.7681425
- M. Terrovitis. (2023). OpenAIRE webinar - Amnesia: High-accuracy Data Anonymization. Zenodo. https://doi.org/10.5281/zenodo.7636541
- Oana Niculaescu. (2017). K-anonymity and cluster-based methods for privacy. https://elf11.github.io/2017/04/22/kanonymity.html
- Vlad and Haaker.(2023). How to anonymise qualitative and quantitative data. ESRC.https://ukdataservice.ac.uk/app/uploads/howtoanonymisequalandquantdata2022-04-21.pdf

Thank you!

Coming up:

May: Metadata – what, why and how?
June: Legal arrangements for RDM

See schedule for more information

Do you have suggestion for a topic?
Contact us via EUR Research Data Stewards